Clinical Risk Assessment Report: Patient 8

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.47x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.8%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.82 [Top 20%] (+1.1%)**

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -1.1 [Bottom 13%] (-2.8%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.94 [Bottom 13%] (-0.9%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.14 [Avg (48th %ile)] (-0.9%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.64 [Bottom 19%] (-0.9%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.62 [Bottom 20%] (-0.8%)**

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.42 [Bottom <1%] (-0.8%)**

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.22 [Avg (45th %ile)] (-0.7%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Machine Learning Feature Attribution (SHAP Decision Path)